

TrES-5b

R-1533

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130519 · created 2026-08-01T18:21:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456431.9097 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.032
Transit depth (Rp/Rs)^2	1.94 +/- 0.89 [%]
Orbital Inclination (inc)	85.69 +/- 2.14
Airmass coefficient 1 (a1)	0.901 +/- 0.02
Airmass coefficient 2 (a2)	0.082 +/- 0.029
Scatter in the residuals of the lightcurve fit is	2.24 %
Best Comparison Star	#3 - [351, 227]
Optimal Aperture	2.81
Optimal Annulus	9.93
Transit Duration (day)	0.0775 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.032 vs 0.142 (deviation 0.09σ)
Duration fit vs published	0.0775 d vs 0.0758 d (deviation 0.15σ)
Mid-time O–C	-3.0 min
Depth SNR	4.0
Residual scatter	2.24 %

Light curve

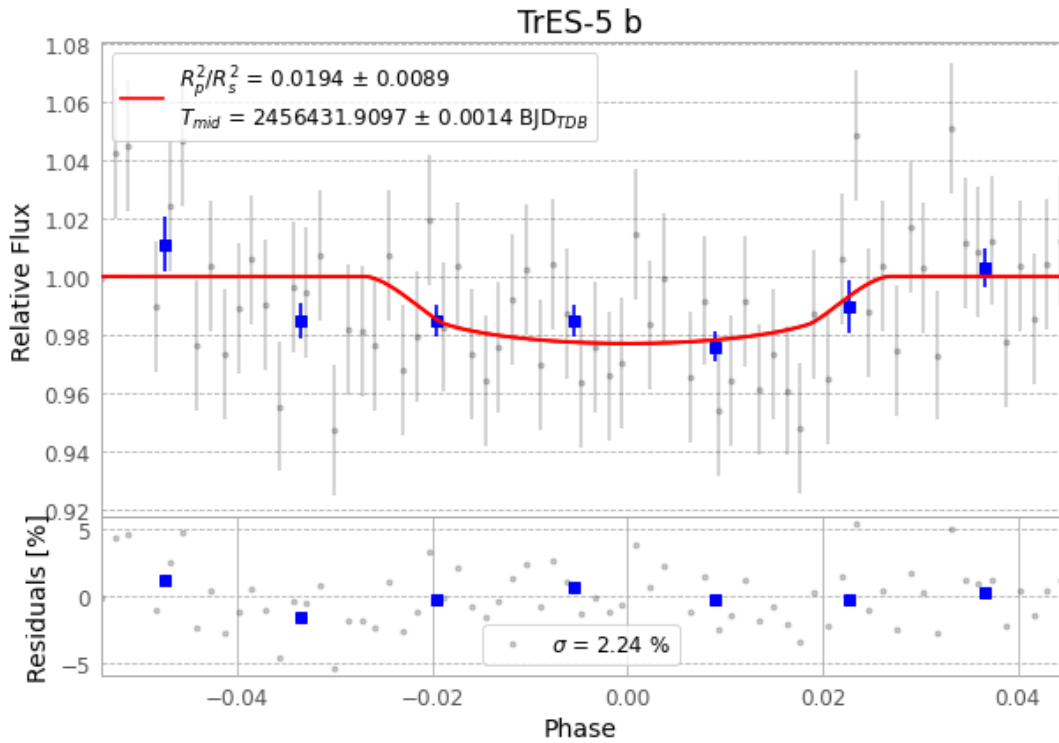


Figure 1 — FinalLightCurve_TrES-5 b_2013-05-19.png (SHA-256 d378a9fe8b0ba595...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3fde30818036501b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-5130519075412.FITS → TRES-5130519112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130519.log (5f8aee3bcf21...), FinalLightCurve_TrES-5 b_2013-05-19.png (d378a9fe8b0b...), FinalLightCurve_TrES-5 b_2013-05-19.csv (cff3f2991657...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:21Z.